Appendix. Robustness of the node-discrimination threshold and the soft-mask parameter

1.1 Scope of the analysis

This appendix works directly from the decomposition results saved for the main text, covering all borehole pairs and both signal types across nine datasets, 1551 decompositions in all: 12 from the four-case TRD dataset used in Sections 2.2.2 and 3.1, 90 from the three RCT groups of Section 3.2, and 1449 from the five lithology datasets used for lithology identification in Section 4.

1.2 Shape of the node-distance distribution and Otsu discrimination

Table 1. Node discrimination for every borehole pair of the four-case TRD dataset.

Signal	Borehole pair	Nodes	Common / differential	Otsu threshold	Largest distance / threshold	η
Pressure	Case-4 – Case-2	32	31 / 1	0.3077	31.4	1.0000
Pressure	Case-4 – Case-3	32	31 / 1	0.2040	31.1	1.0000
Pressure	Case-1 – Case-4	32	31 / 1	0.2136	31.3	1.0000
Pressure	Case-1 – Case-2	32	31 / 1	0.0937	31.9	0.9999
Pressure	Case-1 – Case-3	32	31 / 1	0.0115	30.3	0.9993
Pressure	Case-2 – Case-3	32	31 / 1	0.1044	32.0	1.0000
Torque	Case-4 – Case-2	32	31 / 1	0.4960	29.3	0.9998
Torque	Case-4 – Case-3	32	31 / 1	0.7113	29.9	1.0000
Torque	Case-1 – Case-4	32	31 / 1	0.5386	31.7	0.9999
Torque	Case-1 – Case-2	32	31 / 1	1.0064	31.4	0.9998
Torque	Case-1 – Case-3	32	31 / 1	0.2849	14.8	0.9973
Torque	Case-2 – Case-3	32	31 / 1	1.1198	32.0	0.9999

All 12 give 31 common nodes and one differential node. The largest node distance is 14.8 to 32.0 times the Otsu threshold, and the ratio η of between-class to total variance achieved by the split is 0.9973 to 1.0000. Eleven of the twelve cases fall within the narrow band of 29.3 to 32.0 times, the only outlier being the torque signal of Case-1 – Case-3 at 14.8 times. The Case-1 – Case-2 example shown in Section 2.2.2 of the main text (threshold 0.0937, largest distance 31.9 times the threshold) is a typical member of that band and is therefore not a selectively chosen case with unusually clean separation.

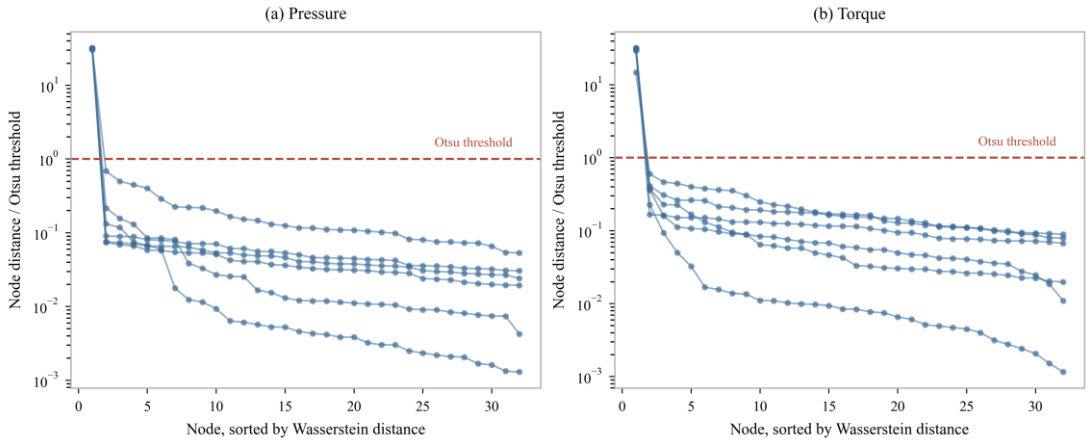


Fig. 1. Node Wasserstein distances for every borehole pair of the four-case TRD dataset (normalized by the Otsu threshold and sorted in descending order).

The 12 curves in Fig. 1 share one shape: the first-ranked node exceeds the threshold by one to two orders of magnitude, and the remaining 31 nodes all fall below it.

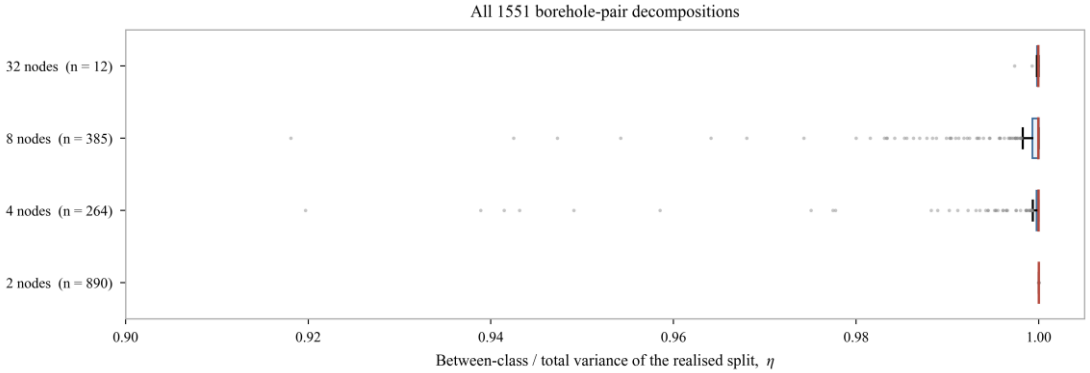


Fig. 2. Split separability η over all 1551 decompositions, stratified by the number of nodes.

Over all 1551 decompositions (Fig. 2), the proportion with exactly one differential node is 100.0% for the 2-, 4- and 32-node cases and 99.7% for the 8-node case, and the median η is 0.9999–1.0000 at every node count. That a single low-frequency node carries almost all of the inter-borehole difference is therefore not a coincidence of particular cases but a structure that holds generally in these data.

1.3 Adaptive choice of the number of levels, and why the separation indices of the RCT dataset are lower

Table 2. Stable-segment length, adaptive number of decomposition levels and number of nodes for each dataset.

Dataset	Decompositions	Stable-segment length (points)	Levels	Nodes
TRD, four cases	12	1720–2043	5	32
RCT group 1	30	53–60	1	2
RCT group 2	30	55–60	1	2
RCT group 3	30	58–60	1	2
Sandstone A	963	10–400	1/2/3	2/4/8
Sandstone B	30	55–60	1	2

Sandstone C	121	55–400	1/2/3	2/4/8
Granite	225	95–400	1/2/3	2/4/8
Limestone	110	50–400	1/2/3	2/4/8

The number of WPD levels is determined adaptively from the signal length, the target being at least 32 samples in each final-level node subject to a maximum of five levels; when the signal is shorter than 64 points that target cannot be met and the decomposition reduces to one level and two nodes. The stable segments of the four-case TRD data are 1720–2043 points long and give five levels and 32 nodes, whereas those of the three RCT groups in Section 3.2 are only 53–60 points and support just one level and two nodes.

This difference explains why the separation indices of the RCT dataset are lower. With only two sub-bands the first-stage coarse decomposition can make no more than the crudest split in frequency, leaving markedly less sub-band structure for the second stage to refine, so the final reduction falls to 78%–82% (Section 3.2 of the main text). This follows from a frequency resolution limited by signal length, not from any instability of the Otsu threshold. With two nodes, any monotone threshold rule can only produce a one-against-one split, and the bimodality assumption plays no part in the discrimination.

1.4 Sensitivity to the node-threshold rule

Table 3. First-stage coarse decomposition under five threshold rules.

Dataset	Threshold rule	Common nodes	Coarse CC reduction (%)	Coarse DiC retention (%)	Degenerate cases
TRD, four cases	Otsu	31	97.95	100.00	0
	Median	16	99.37	100.00	0
	Mean	31	97.95	100.00	0
	75th percentile	24	98.96	100.00	0
	90th percentile	28	98.56	99.99	0
RCT group 1	Otsu	1	96.31	100.07	0
	Median	1	96.31	100.07	0
	Mean	1	96.31	100.07	0
	75th percentile	1	96.31	100.07	0
	90th percentile	1	96.31	100.07	0
RCT group 2	Otsu	1	94.69	100.04	0
	Median	1	94.69	100.04	0
	Mean	1	94.69	100.04	0
	75th percentile	1	94.69	100.04	0
	90th percentile	1	94.69	100.04	0
RCT group 3	Otsu	1	96.72	100.12	0
	Median	1	96.72	100.12	0
	Mean	1	96.72	100.12	0
	75th percentile	1	96.72	100.12	0
	90th percentile	1	96.72	100.12	0

On the four-case TRD data the coarse CC reduction of the five rules is 97.95%–99.37% and the coarse DiC retention is never below 99.99%, with no degenerate case (all nodes assigned to one class); Otsu and the mean give exactly the same partition. On the three RCT groups the five rules

give identical results, for the reason given above: with two nodes any threshold can only split one against one. The coarse DiC retention of those three groups slightly exceeds 100% (100.04%–100.12%) because the Wasserstein distance is not additive over a decomposition, so once the common component is removed the residuals can lie slightly farther apart than the original signals; this does not affect the conclusions on separation.

1.5 Sensitivity to the Sigmoid soft mask

The Sigmoid mask takes the mean of the frequency-wise Wasserstein distance as its inflection point and the standard deviation multiplied by a width coefficient σ as its scale. Table 4 and Fig. 3 give the results of the complete two-stage workflow for σ from 0.5 to 2.0.

Table 4. Sensitivity to the Sigmoid width coefficient σ (complete two-stage workflow, mean ± standard deviation).

σ	TRD CC reduction (%)	TRD DiC retention (%)	RCT CC reduction (%)	RCT DiC retention (%)
0.50	97.56 ± 4.05	98.41 ± 3.68	89.21 ± 12.50	89.87 ± 12.85
0.75	95.37 ± 5.92	95.78 ± 5.79	83.46 ± 12.05	83.73 ± 12.30
1.00 (used in the main text)	92.12 ± 6.90	92.27 ± 6.79	78.48 ± 11.35	78.60 ± 11.49
1.50	84.97 ± 7.14	84.99 ± 7.08	71.34 ± 9.60	71.39 ± 9.67
2.00	79.00 ± 6.58	79.01 ± 6.55	66.84 ± 8.07	66.86 ± 8.10

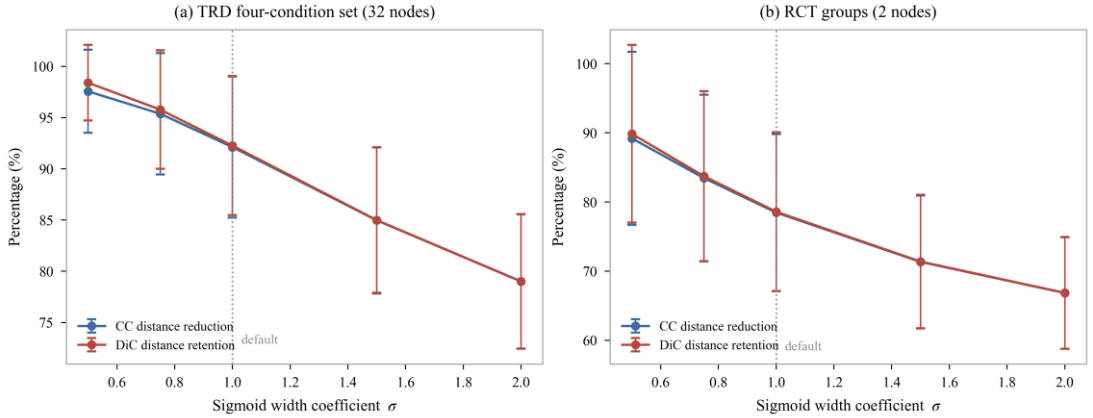


Fig. 3. Effect of the Sigmoid width coefficient σ on the final separation indices.

A smaller σ brings the mask closer to a hard threshold, and the CC reduction and DiC retention rise together (four-case TRD data: 97.6%/98.4% at σ = 0.5, 92.1%/92.3% at σ = 1.0 and 79.0%/79.0% at σ = 2.0). That the indices improve monotonically as σ decreases does not, however, mean that a smaller σ should be adopted. The purpose of the soft mask is precisely to replace a hard-threshold split with continuous routing so as to avoid Gibbs oscillation (Section 2.3.2 of the main text), and σ = 1.0 is a compromise between routing smoothness and separation. More importantly, the qualitative conclusion that the common component converges while the differential component diverges holds across the whole range of σ: even in the least favourable case, σ = 2.0, both indices remain near 79%.

Table 5. Comparison of three mask modes (σ = 1.0, four-case TRD dataset).

Mask mode	CC reduction (%)	DiC retention (%)
Sigmoid (used in the main text)	92.12	92.27

Otsu soft mask	86.23	86.82
Linear normalization	81.45	82.11

Sigmoid outperforms the other two modes on both indices, supporting the choice made in the main text.

1.6 Conclusions

(1) That a single low-frequency node carries almost all of the inter-borehole difference holds generally across all 1551 decompositions counted here, so the case used in the main text is representative rather than selectively displayed. (2) The first-stage discrimination is insensitive to the threshold rule, and the lower separation indices of the RCT dataset arise from a frequency resolution limited by stable-segment length, not from instability of the Otsu threshold. (3) The value of σ affects the numerical degree of separation but does not change the qualitative conclusion that the common component converges while the differential component diverges.